INTERNNOVA - JAVA INTERNSHIP

WEEK 2 - ASSIGNMENT

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College: PSNA College of Engineering and Technology

Internship Domain: Java Development

Assignment: Week 2 – Control Flow and Methods

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OBJECTIVE & LEARNING OUTCOMES

The main objective of this assignment is to strengthen the understanding of fundamental Java programming concepts through practical implementation. The assignment provides hands-on experience in developing Java programs using different programming constructs and problem-solving techniques.

This assignment focuses on important concepts such as conditional statements, looping statements, methods, method parameters, return types, one-dimensional arrays, and two-dimensional arrays. By implementing individual programs for each concept, the basic syntax and working principles of Java can be understood more clearly.

The programs are designed to accept input from the user, process the given data, and produce the required output. This helps in understanding how Java programs interact with users and how different programming concepts can be combined to solve simple problems.

Another important objective of this assignment is to improve logical thinking and programming skills. By using conditions, loops, methods, and arrays, different problems can be solved in a structured and efficient manner.

The assignment also provides practical experience in writing, compiling, debugging, and executing Java programs. This helps in identifying errors, improving code quality, and developing confidence in Java programming.

TASK 1: CONDITIONAL STATEMENTS – STUDENT RESULT

Objective

To create a Java program that accepts the marks of a student in three subjects, calculates the total marks and percentage, and determines the student's grade and result using conditional statements.

Brief Explanation

This program accepts the student's name and marks in three subjects as input. It calculates the total marks and percentage using arithmetic operations. Conditional statements such as if, else if, and else are used to determine the student's grade and result based on the calculated percentage.

Source Code

File Name: StudentResult.java

```
import java.util.Scanner;
public class StudentResult {
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println("Student Result");
        System.out.println("--------------");
```

```java
System.out.println();
System.out.print("Enter Student Name : ");
String name=sc.nextLine();
System.out.println();
System.out.println("Marks");
System.out.println("-----");
System.out.print("Subject 1 : ");
int S1=sc.nextInt();
System.out.print("Subject 2 : ");
int S2=sc.nextInt();
System.out.print("Subject 3 : ");
int S3=sc.nextInt();
System.out.println();
System.out.println("Result");
System.out.println("------");
System.out.println("Total Marks : "+(S1+S2+S3));
float Percentage=(S1+S2+S3)/3.0f;
System.out.printf("Percentage : %.2f%%",Percentage);
System.out.println();
```

```
        System.out.print("Grade : ");
        if(Percentage>=90)
            System.out.println("A+");
        else if(Percentage>=80)
            System.out.println("A");
        else if(Percentage>=70)
            System.out.println("B");
        else if (Percentage>=60)
            System.out.println("C");
        else if (Percentage>=50)
            System.out.println("D");
        else
            System.out.println("F");
        System.out.print("Result : ");
        if(Percentage>=50)
            System.out.print("Pass");
        else
            System.out.print("Fail");
    }
}
```

Output

```
PS D:\Internova\Week_2\Task_1> java StudentResult.java
Student Result
--------------

Enter Student Name : Subhasri S

Marks
-----
Subject 1 : 90
Subject 2 : 95
Subject 3 : 91

Result
------
Total Marks : 276
Percentage : 92.00%
Grade : A+
Result : Pass
```

TASK 2: LOOPS – NUMBER PRACTICE

Objective

To use different types of loops in Java to print numbers, identify even and odd numbers, and calculate the sum of numbers from 1 to 100.

Brief Explanation

This program demonstrates the use of looping statements in Java. It prints numbers from 1 to 100, displays all even and odd numbers, and calculates the sum of numbers from 1 to 100. Different loops such as while, for, and do-while are used to perform these operations.

Source Code

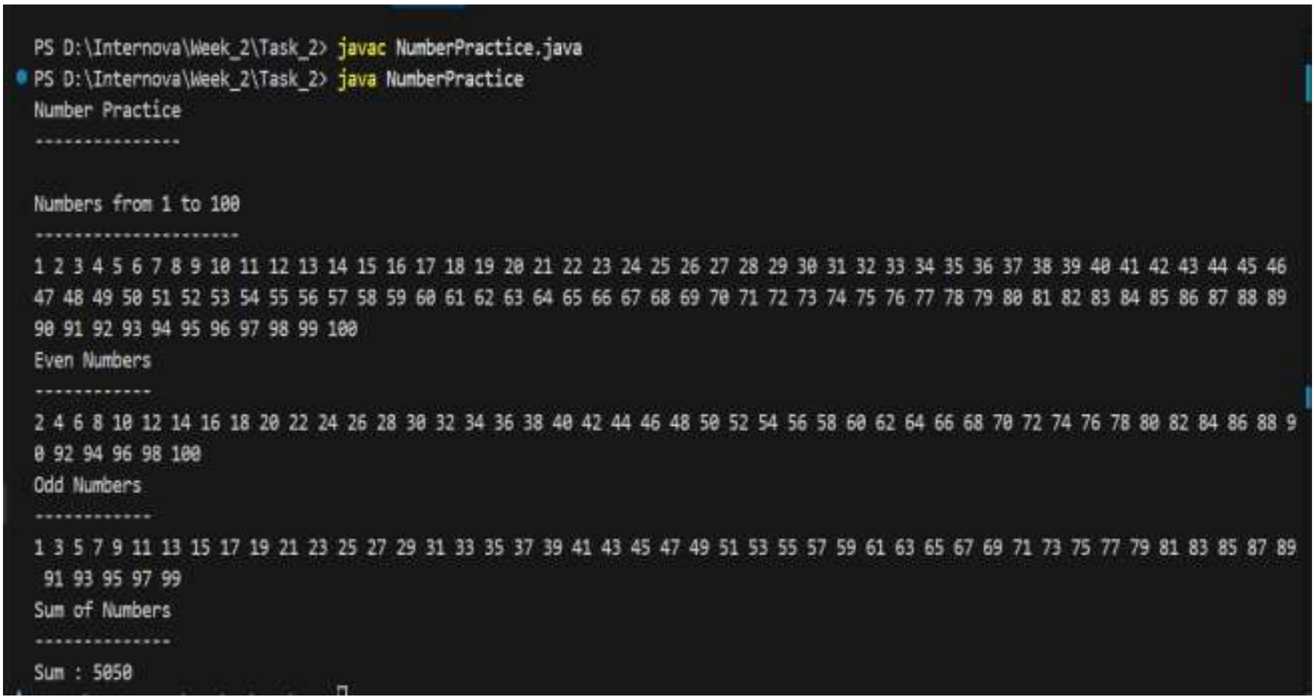
File Name: NumberPractice.java

```
public class NumberPractice {
    public static void main(String[] args) {
        System.out.println("Number Practice");
        System.out.println("---------------");
        System.out.println();
        System.out.println("Numbers from 1 to 100");
        System.out.println("---------------------");
```

```
int i=1;
while(i<=100){
    System.out.print(i+" ");
    i++;
}
System.out.println();
System.out.println("Even Numbers");
System.out.println("------------");
for(i=1;i<=100;i++){
    if(i%2==0)
        System.out.print(i+" ");
}
System.out.println();
System.out.println("Odd Numbers");
System.out.println("------------");
for(i=1;i<=100;i++){
    if(i%2!=0)
        System.out.print(i+" ");
}
System.out.println();
```

```
        System.out.println("Sum of Numbers");
        System.out.println("--------------");
        int sum=0;
        i=1;
        do{
            sum+=i;
            i++;
        }while(i<=100);
        System.out.println("Sum : "+sum);
    }
}
```

Output


```
PS D:\Internova\Week_2\Task_2> javac NumberPractice.java
PS D:\Internova\Week_2\Task_2> java NumberPractice
Number Practice
---------------

Numbers from 1 to 100
--------------------
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46
47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99 100
Even Numbers
------------
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 9
0 92 94 96 98 100
Odd Numbers
------------
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89
 91 93 95 97 99
Sum of Numbers
--------------
Sum : 5050
```

TASK 3: METHODS – CALCULATOR

Objective

To create a calculator program using separate methods for performing basic arithmetic operations.

Brief Explanation

This program demonstrates the use of methods in Java by creating separate methods for addition, subtraction, multiplication, division, and modulus operations. Two numbers are accepted from the user, and the appropriate methods are called to calculate and display the results.

Source Code

File Name: Calculator.java

```
import java.util.Scanner;
public class Calculator {
  public static void main(String[] args) {
      Scanner sc=new Scanner(System.in);
      Calculator ca=new Calculator();
      System.out.println("Method Calculator");
      System.out.println("----------------");
      System.out.println();
      System.out.print("Enter first Number  : ");
```

```
        int num1=sc.nextInt();
        System.out.print("Enter Second Number : ");
        int num2=sc.nextInt();
        System.out.println();
        System.out.println("Results");
        System.out.println("-------");
        int c=ca.add(num1,num2);
        System.out.println("Addition          : "+c);
        System.out.print("Subtraction       : ");
        ca.sub(num1,num2);
        System.out.print("Multiplication    : ");
        ca.mul(num1,num2);
        float d=ca.div(num1,num2);
        System.out.printf("Division          : %.2f",d);
        System.out.println();
        System.out.print("Modulus           : ");
        ca.mod(num1,num2);
    }
  int add(int a,int b){
   return a+b;
```

```
}
void sub(int a,int b){
 System.out.println(a-b);
}
void mul(int a,int b){
 System.out.println(a*b);
}
float div(float a,float b){
 if(b==0){
     System.out.println("Divisor Cannot be zero");
     return 0.0f;
 }
 float c= a/b;
 return c;
}
void mod(int a,int b){
  if(b==0){
     System.out.println("Divisor Cannot be zero");
     return;
 }
```

```
    System.out.println(a%b);
  }
}
```

Output

```
PS D:\Internova\Week_2\Task_3> javac Calculator.java
PS D:\Internova\Week_2\Task_3> java Calculator
Method Calculator
-----------------

Enter first Number  : 20
Enter Second Number : 6

Results
-------
Addition          : 26
Subtraction       : 14
Multiplication    : 120
Division          : 3.33
Modulus           : 2
```

TASK 4: METHOD PARAMETERS & RETURN TYPES

Objective

To create methods that accept parameters and return appropriate results for different mathematical operations.

Brief Explanation

This program demonstrates the use of method parameters and return types in Java. Separate methods are created to find the square and cube of a number, calculate the average of three numbers, and find the maximum of two numbers. Each method accepts the required input as parameters and returns the calculated result.

Source Code

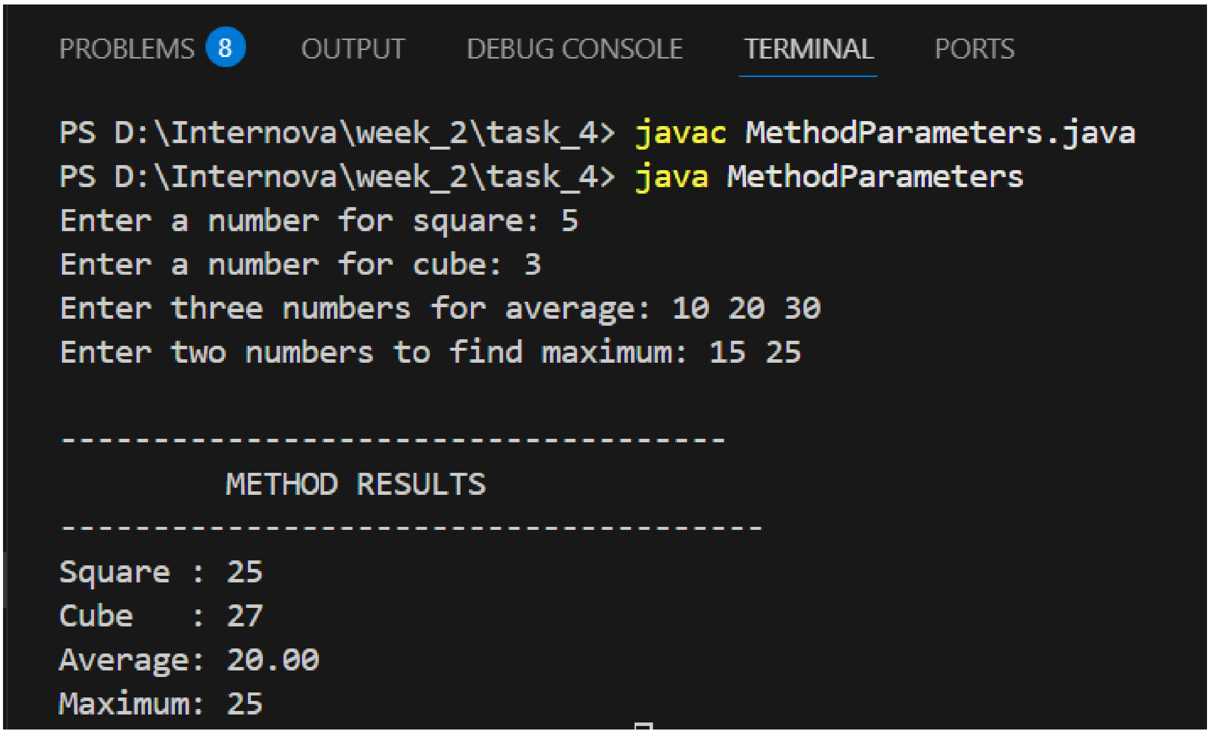
File Name: MethodParameters.java

```
import java.util.Scanner;
public class MethodParameters {
    static int square(int n) {
        return n * n;
    }
    static int cube(int n) {
        return n * n * n;
    }
```

```
    static float average(float a, float b, float c) {
        return (a + b + c) / 3.0f;
    }
    static int maximum(int a, int b) {
        return (a > b) ? a : b;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number for square: ");
        int n = sc.nextInt();
        System.out.print("Enter a number for cube: ");
        int num = sc.nextInt();
        System.out.print("Enter three numbers for average:
");
        float a = sc.nextFloat();
        float b = sc.nextFloat();
        float c = sc.nextFloat();
        System.out.print("Enter two numbers to find
maximum: ");
        int x = sc.nextInt();
        int y = sc.nextInt();
        System.out.println("\n=====================
===================");
        System.out.println("        METHOD RESULTS");
        System.out.println("======================
```

```
==================");
        System.out.println("Square : " + square(n));
        System.out.println("Cube   : " + cube(num));
        System.out.printf("Average: %.2f%n", average(a,
b, c));
        System.out.println("Maximum: " + maximum(x,
y));
        sc.close();
    }
}
```

Output



TASK 5: 1D ARRAY – STUDENT MARKS

Objective

To use a one-dimensional array to store the marks of five students and calculate their total, average, highest, and lowest marks.

Brief Explanation

This program demonstrates the use of a one-dimensional array in Java. It accepts the marks of five students from the user and stores them in an array. The program then displays all the marks and calculates the total, average, highest, and lowest marks using loops and array elements.

Source Code

File Name: StudentMarks.java

```
import java.util.Scanner;
public class StudentMarks {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] marks = new int[5];
        int total = 0;
        System.out.println("Enter marks of 5 students:");
        for (int i = 0; i < marks.length; i++) {
            System.out.print("Student " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
```

```
            total += marks[i];
        }
        int highest = marks[0];
        int lowest = marks[0];
        for (int i = 1; i < marks.length; i++) {
            if (marks[i] > highest) {
                highest = marks[i];
            }
            if (marks[i] < lowest) {
                lowest = marks[i];
            }
        }
        double average = total / 5.0;
        System.out.println("\n=======================
======================");
        System.out.println("          STUDENT MARKS");
        System.out.println("==========================
===================");
        System.out.print("Marks: ");
        for (int mark : marks) {
            System.out.print(mark + " ");
        }
        System.out.println("\nTotal Marks   : " + total);
        System.out.printf("Average Marks : %.2f%n",
average);
        System.out.println("Highest Marks : " + highest);
```

```
        System.out.println("Lowest Marks  : " + lowest);
        sc.close();
    }
}
```

Output

```
PS D:\Internova\week_2\task_5> javac StudentMarks.java
PS D:\Internova\week_2\task_5> java StudentMarks
Enter marks of 5 students:
Student 1: 85
Student 2: 92
Student 3: 78
Student 4: 88
Student 5: 95

======================================
          STUDENT MARKS
======================================
Marks: 85 92 78 88 95
Total Marks    : 438
Average Marks : 87.60
Highest Marks : 95
Lowest Marks  : 78
```

TASK 6: 2D ARRAY – MATRIX OPERATIONS

Objective

To use a two-dimensional array to store and display a 3 × 3 matrix and calculate the sum of all its elements.

Brief Explanation

This program demonstrates the use of a two-dimensional array in Java. It accepts nine elements from the user and stores them in a 3 × 3 matrix. Nested loops are used to access and display the matrix elements. The program also calculates and displays the sum of all the elements in the matrix.

Source Code

File Name: MatrixOperations.java

```
import java.util.Scanner;
public class MatrixOperations {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][] matrix = new int[3][3];
        int sum = 0;
        System.out.println("Enter elements of 3x3
matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print("Enter element [" + i + "]["
```

```
+ j + "]: ");
                matrix[i][j] = sc.nextInt();

                sum += matrix[i][j];
            }
        }
        System.out.println("\n======================
====================");
        System.out.println("             MATRIX");
        System.out.println("========================
==================");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(matrix[i][j] + "\t");
            }
            System.out.println();
        }
        System.out.println("\nSum of all elements: " +
sum);
        sc.close();
    }
}
```

Output

```
PS D:\Internova\week_2\task_6> javac MatrixOperations.java
PS D:\Internova\week_2\task_6> java MatrixOperations
Enter elements of 3x3 matrix:
Enter element [0][0]: 1
Enter element [0][1]: 2
Enter element [0][2]: 3
Enter element [1][0]: 4
Enter element [1][1]: 5
Enter element [1][2]: 6
Enter element [2][0]: 7
Enter element [2][1]: 8
Enter element [2][2]: 9

======================================
             MATRIX
======================================
1       2       3
4       5       6
7       8       9

Sum of all elements: 45
```